A Triple-Branch Hybrid Attention Network With Bitemporal Feature Joint Refinement for Remote-Sensing Image Semantic Change Detection

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***Abstract*— Compared with binary change detection (BCD), semantic change detection (SCD) further provides the category information of bitemporal changed regions which is significant for the practical application of Earth observation. Although the recently proposed triple-branch structures including one BCD branch and two classification branches can effectively achieve task balance, they still need to employ the carefully designed difference extraction (DE) module and branch interactions to capture the bitemporal correlations, which increases the complexity of the semantic information utilization. In this article, we propose a new triple-branch network named JFRNet to tackle this challenge. From the perspective of the SCD process, because the category information and the change information are both derived from bitemporal images, we take the joint bitemporal features as the unified input, which can help each branch perceive the bitemporal semantic correlations without any additional interaction operations. From the perspective of the SCD structure, we introduce the convolutional attention fusion module (CAFM) and the convolutional attention refinement module (CARM) to unify the branch structure, which can help our model refine the unique semantic information without any specially designed DE modules. Extensive experiment results on three available datasets indicate that compared with the baseline methods, our proposed JFRNet successfully simplifies the reasoning process and obtains better SCD performance.**

***Index Terms*— Attention mechanism, branch interaction, convolutional neural network (CNN), semantic change detection (SCD).**

I. Introduction

A. Background

GIVEN bitemporal images, change detection (CD) aims to automatically identify the change information of interest ground objects, which can reduce the manual statistics and improve the interpretation efficiency [1], [2], [3], [4], [5], [6]. In recent years, with the powerful projection ability, deep-learning CD methods have achieved remarkable results in binary CD (BCD) task [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21]. Many advanced algorithms can obtain precise results on various available BCD datasets [8], [22], [23], [24]. However, as shown in Fig. 1, BCD only can recognize whether ground objects have changed, but cannot distinguish the categories, which limits the application of CD in some fine-grained tasks, such as land-cover and land-use [25], [26]. As a result, semantic CD (SCD) received increasing attention recently.

Because SCD needs to perform both the change and the segmentation tasks, many studies try to achieve improvements in the branch structure. Currently, the feasible SCD models can be divided into three parts: single-branch structure, dual-branch structure, and triple-branch structure. The single-branch structure directly adopts one difference branch to obtain the semantic change information [27]. However, this strategy generally needs to transfer bitemporal semantic change maps (SCMs) into one category change map, which makes models need to consider all of the "from–to" semantic category transitions. As a result, if the changed regions contain N types of ground objects, the model needs to totally consider $N(N-1)$ types of category change combinations which will badly increase the prediction difficulty. The dual-branch structure usually sets two classification branches to directly obtain bitemporal SCMs [28], [29]. However, because SCM still represents the change information, this strategy generally needs to introduce some branch interactions to perceive the bitemporal differences, which makes models difficult to keep a good balance between the change and the segmentation tasks.

Differently, the recently proposed triple-branch structure utilizes one BCD branch and two classification branches to, respectively, predict BCM and bitemporal category maps, and further masks bitemporal category maps with BCM to generate SCMs [30], [31], [32]. Compared with the other two feasible structures mentioned above, this strategy can balance the change and the segmentation tasks more effectively. As a result, the triple-branch structure has received increasing attention recently.

B. Challenge and Motivation

However, the existing triple-branch methods still have two key challenges in the utilization of bitemporal correlations,

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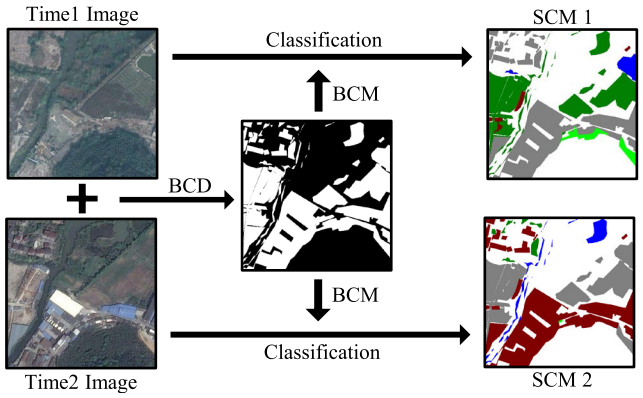


Fig. 1. Illustration of BCD and SCD. Based on the BCM, SCD further segments the bitemporal changed regions to obtain the SCMs.

which include two aspects: semantic information difference and semantic information consistency.

1) Semantic Difference: As shown in Fig. 2(a), because BCM directly determines the boundaries of changed regions, the existing methods [30], [32], [33] commonly need to introduce the carefully designed DE module to obtain the semantic difference between bitemporal features. For example, Niu et al. [32] introduce a multicontent module to obtain abundant change information. Ding et al. [30] adopt a convolution block with six stacked residual units [34] to achieve DE. This is because the preliminary DE module (such as shallow convolution layers) makes it difficult to extract the precise difference information. Therefore, how to design an effective DE module is a key challenge that the existing triple-branch methods need to face.

2) Semantic Consistency: Although bitemporal images can contain certain changes over time, they essentially still represent the ground information in the same area. As a result, the bitemporal features of the same objects still have high semantic consistency which can help models better understand the ground categories. Therefore, as shown in Fig. 2(a), many studies [30], [31], [33] also introduce the additional branch interactions to connect bitemporal image features. For example, Yang et al. [33] introduce a specially designed U-shape connection module to achieve feature interaction. Ding et al. [30] utilize self-attention and cross-attention [35] to refine bitemporal features. However, these interaction strategies are usually across the semantic feature spaces, such as bitemporal feature space and difference feature space. Therefore, they generally require a higher long-distance modeling ability to capture the cross-feature correlations. As a result, how to make sure the introduced branch interactions are effective is another key challenge that the existing triple-branch methods need to face.

In this article, we try to propose a new triple-branch network that can make full use of the bitemporal information while avoiding the need for both the carefully designed DE module and branch interactions.

C. Solution and Contribution

Fig. 2(b) illustrates our solution strategy that can be divided into two aspects: the SCD process and the SCD structure. From the perspective of the SCD process, considering the category information and the change information are both derived from bitemporal images, we take the joint bitemporal image features as the unified input, which can help each branch perceive the bitemporal semantic correlations without any additional interaction operations. From the perspective of the SCD structure, because each branch shares the bitemporal information, we unify the difference branch and bitemporal classification branches into three independent refinement branches which are used to extract the unique semantic information for different tasks. As a result, we can also avoid the need for the carefully designed DE module.

Fig. 3 displays the detail process of our proposed refinement branch. As shown in Fig. 3(a), although Chen et al. [14] have successfully deployed the attention mechanism to capture bitemporal correlations, the image features after the refinement are still independent of each other. As a result, this strategy essentially still requires DE operation to obtain the change information. Peculiarly, because the refined features usually represent high-level information, the simple subtraction operation may lose the key semantic category information of bitemporal images. However, in our proposed strategy, we first divide the refinement branch into two stages: bitemporal feature fusion and semantic information refinement. Then, we propose the convolutional attention fusion module (CAFM) and the convolutional attention refinement module (CARM) to perform the whole process, which can better integrate the image features and tackle both the difference and segmentation tasks.

To sum up, the contributions of this study can be summarized as the following four points.

1) We propose a triple-branch network named JFRNet for remote-sensing image SCD. Compared with the existing methods, we provide a new research perspective for SCD: converting the multitask independent learning into bitemporal feature joint refinement.
2) From the perspective of the SCD process, we take the joint bitemporal image features as the unified input, which can help each branch perceive the bitemporal semantic correlations without any additional interaction operations.
3) From the perspective of the SCD structure, we introduce the CAFM and the CARM to unify the branch structure, which can help our model extract precise semantic information without any specially designed DE blocks.
4) Extensive experiment results on three available datasets indicate that compared with the baseline methods, our proposed JFRNet successfully simplifies the reasoning process and obtains better SCD performance.

In Sections I–VI, we first analyze the existing BCD and SCD methods. After that, we introduce our proposed method and the experimental results in detail. Finally, we discuss the model deficiency and give the conclusion.

II. Related Work

According to the prediction task, the existing CD methods can be mainly divided into two parts: BCD and SCD.

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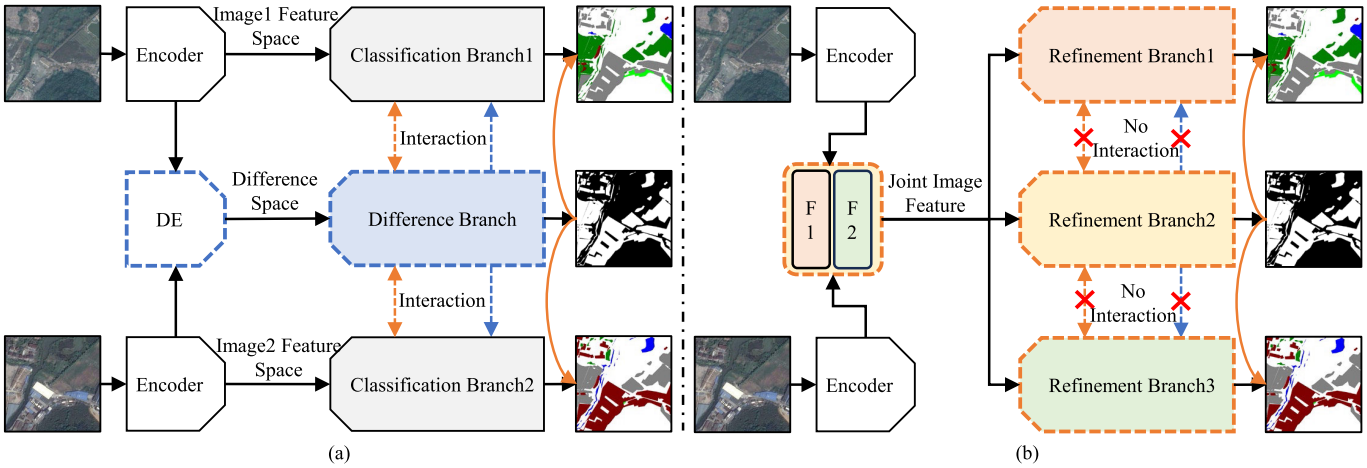


Fig. 2. Structure comparison between (b) our proposed strategy and (a) existing refinement strategy. The red and blue dashed arrows represent the branch interaction operations. The red solid arrows represent the masking operation. (a) Existing strategy generally needs to deploy the specially designed DE module and branch interactions to perform SCD. However, in (b) our proposed strategy, we take the joint bitemporal image features as the common input and further unify the branch structure to simplify the complexity of the semantic information utilization.

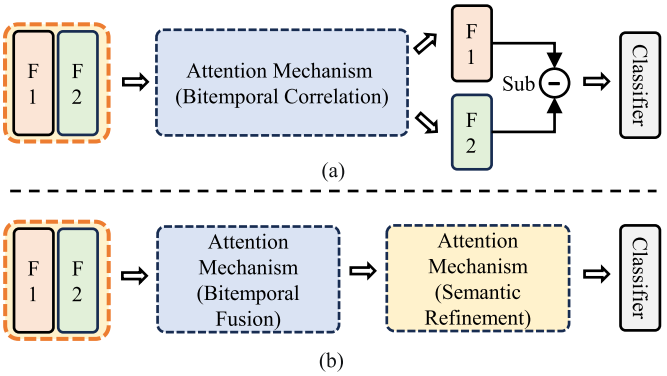


Fig. 3. Branch process comparison between (b) our proposed strategy and (a) existing strategy. Although (a) strategy has successfully deployed the attention mechanism to capture bitemporal correlations, the image features after the refinement are still independent of each other. As a result, this strategy also adopts the subtraction (sub) operation to directly obtain the change information, which may lose the key semantic category information of bitemporal images. However, in (b) our proposed strategy, we divide the refinement branch into two stages: bitemporal feature fusion and semantic information refinement, which can better integrate the image features and tackle both the difference and segmentation tasks.

A. BCD Methods

As shown in Fig. 1, given bitemporal images, BCD outputs the binary change map (BCM), where "1" represents changed and "0" represents unchanged. Therefore, the BCD process can be expressed as

$$\mathrm{F_{BCD}}\left(\mathrm{img}_1^{i,j}, \mathrm{img}_2^{i,j}\right) = \begin{cases} 0, c_1^{i,j} = c_2^{i,j} \\ 1, c_1^{i,j} \neq c_2^{i,j} \end{cases} \tag{1}$$

where $\mathrm{F_{BCD}}$ represents the BCD network, $(\mathrm{img}_1^{i,j}, \mathrm{img}_2^{i,j})$ represent the pixel values of position (i, j) in pretime image and posttime image, respectively, and $(c_1^{i,j}, c_2^{i,j})$ represent the serial numbers of position (i, j) in the pretime category map and posttime category map, respectively.

BCD generally contains two key stages: bitemporal feature extraction and difference discrimination. Specifically, both the precise image feature and difference feature can help models perceive the changed regions more easily. Therefore, many BCD methods focus on the improvement of these two aspects. For example, FC-SC [7], FC-SD [7], and FC-EF [7] integrate the multiscale feature fusion strategy into these two stages by modifying UNet [36]. STANet [22] utilizes the self-attention mechanism to refine the bitemporal image features. IFNet [8] introduces the channel and spatial attention to refine the difference feature. BIT [14] utilizes the transformer [37] to capture the spatiotemporal correlations of bitemporal images. DiFormer [19] employs Swin Transformer [38], [39] to achieve the difference enhancement.

However, as shown in Fig. 1, BCD can recognize only whether the ground objects have changed, but cannot distinguish their categories, which limits the application of CD in some fine-grained tasks such as land cover and land use [25], [26], [40].

B. SCD Methods

As shown in Fig. 1, compared with BCD, SCD further segments the changed regions. Therefore, the output of SCD can be unified into the bitemporal semantic change maps (SCMs), where "0" still represents unchanged and the other values represent the specific categories of changed objects

$$\mathrm{F_{SCD}}\left(\mathrm{img}_1^{i,j}, \mathrm{img}_2^{i,j}\right) = \begin{cases} (0,0), c_1^{i,j} = c_2^{i,j} \\ \left(c_1^{i,j}, c_2^{i,j}\right), c_1^{i,j} \neq c_2^{i,j}. \end{cases} \tag{2}$$

$\mathrm{F_{SCD}}$ represents the SCD network.

Currently, one of the feasible strategies is directly modifying the classifier of the BCD network. For example, Pyramid-SCDFormer [27] first utilizes the hierarchical transformer to extract bitemporal image features and then introduces the specially designed difference branch to obtain the result. However, as mentioned in Section I, this strategy generally needs to transfer bitemporal SCMs into one category change map, which makes models need to consider all of the "from–to" semantic category transitions. As a result, the prediction difficulty will be greatly increased. The other feasible structure

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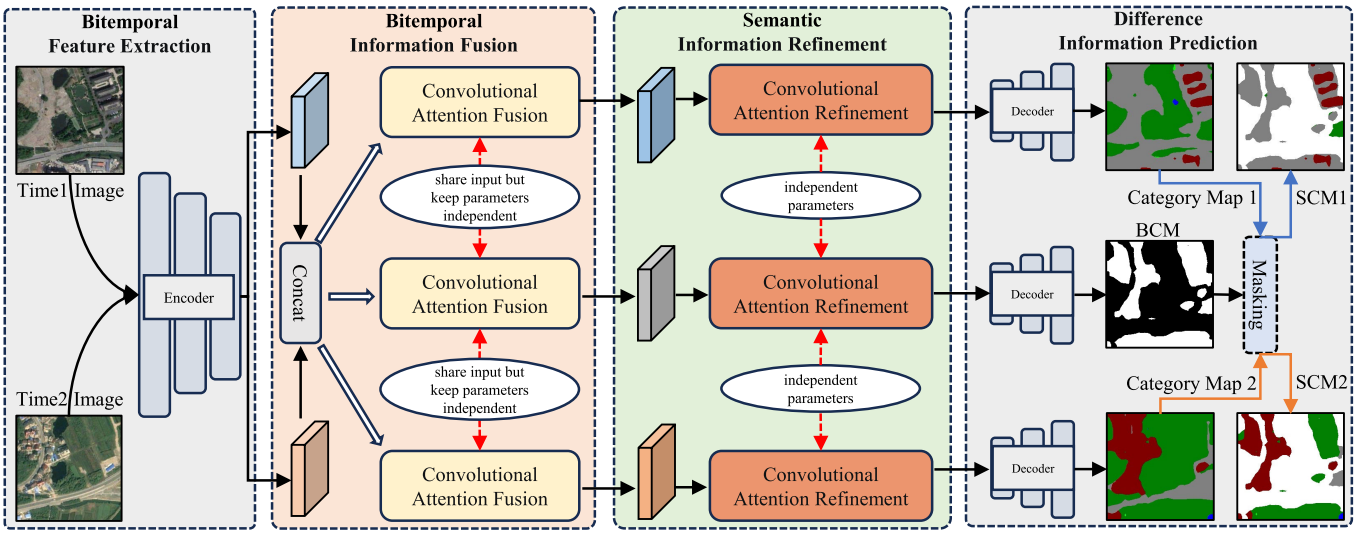


Fig. 4. Illustration of our proposed JFRNet. We first utilize a CNN encoder to extract bitemporal image features. Then, we concatenate and send them into three structurally unified branches which deploy CAFM and CARM to, respectively, perform bitemporal information fusion and semantic information refinement. Finally, we adopt three shallow decoders to obtain the bitemporal category maps and BCM and further mask the category maps with BCM to generate SCMs.

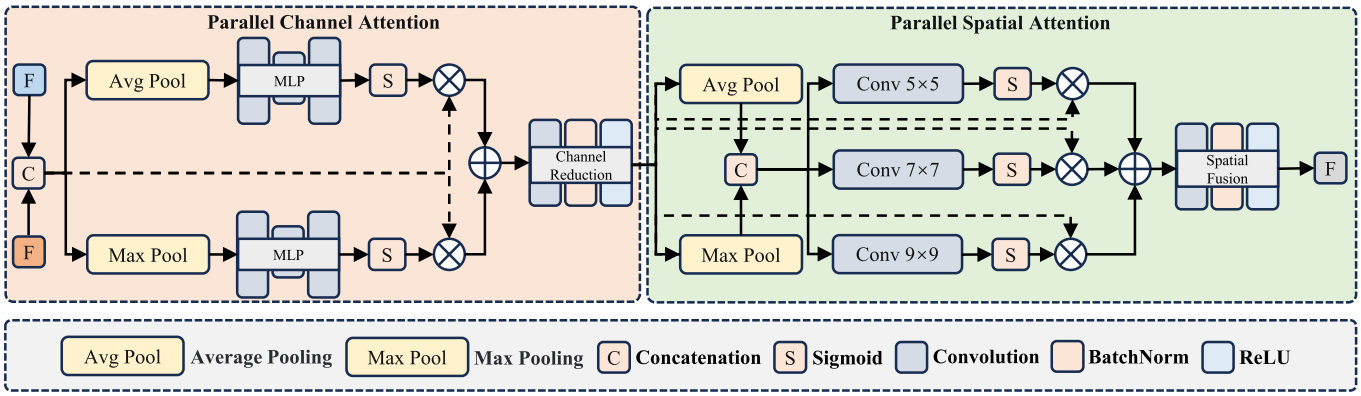


Fig. 5. Illustration of CAFM. We propose the parallel channel attention module and parallel spatial attention module to construct it which set parallel subbranches to capture the attention information and further adopt the channel reduction and spatial fusion operations to update the incorporated features.

is setting two classification branches to directly predict the bitemporal SCMs. For example, PCFN [28] and SCDNet [29] adopt two UNet-style branches that introduce the convolutional attention mechanism to predict SCMs. However, this strategy generally needs some branch interaction operations to perceive the bitemporal differences, which increases the difficulty of the task balance.

Recently, the triple-branch structure has received increasing attention. This strategy utilizes one BCD branch and two classification branches to, respectively, predict BCM and bitemporal category maps and further masks bitemporal category maps with BCM to generate SCMs. Compared with the other two feasible models mentioned above, this strategy can effectively achieve task balance and obtain more precise results. For example, SMNet [32] set three UNet-style branches to fully utilize the multiscale information. HRSCD-str3 [31] integrates the residual blocks [34] into the UNet branches to enhance the feature extraction. However, as shown in Fig. 2(a), the existing triple-branch methods [30], [31], [32], [33] generally need to employ the carefully designed DE module and branch interactions to capture the bitemporal correlations, which increases the complexity of semantic information utilization. Therefore, how to explore an effective SCD process is still worth more exploration.

III. Methodology

As shown in Fig. 4, we first adopt a CNN encoder to extract the bitemporal image features. Then, we concatenate and send them into three structurally unified branches to refine the specific semantic information. For each branch, it consists of two modules: CAFM and CARM. CAFM is used to achieve the preliminary fusion of bitemporal features, and CARM is used to capture and enhance the more salient semantic information. Finally, we utilize three shallow decoders to, respectively, predict the bitemporal category maps and BCM and further mask bitemporal category maps with BCM to obtain the SCMs. Algorithm 1 shows the whole process of our method.

A. Encoder

Following the previous work [30], we adopt the modified ResNet34 [34] as our encoder which outputs the 8× downsampled image features. The process of bitemporal feature

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Algorithm 1 Process of JFRNet for SCD

Input: Bitemporal images: Img_{time1}, Img_{time2}

Output: Bitemporal SCMs: SCM_{pre}, SCM_{post}

// Extracting bitemporal image features by the encoder

$F_{pre} = \text{Encoder}(Img_{time1})$

$F_{post} = \text{Encoder}(Img_{time2})$

// Concatenating bitemporal image features as the unified input

$F_{joint} = \text{Cat}(F_{pre}, F_{post})$

// Utilizing CAFM to achieve the bitemporal information fusion

for $i = 1$ *to* 3 **do**

  $F^i_{CAFM} = \text{CAFM}_i(F_{joint})$

end

// Utilizing CARM to achieve the semantic information refinement

for $i = 1$ *to* 3 **do**

  $F^i_{CARM} = \text{CARM}_i(F^i_{CAFM})$

end

// Utilizing three decoders to obtain the bitemporal category maps and BCM

$Catg_{pre} = \text{Decoder}_1(F^1_{CARM})$

$BCM = \text{Decoder}_2(F^2_{CARM})$

$Catg_{post} = \text{Decoder}_3(F^3_{CARM})$

//Masking bitemporal category maps with BCM to obtain SCMs

$SCM_{pre} = Catg_{pre} * BCM$

$SCM_{post} = Catg_{post} * BCM$

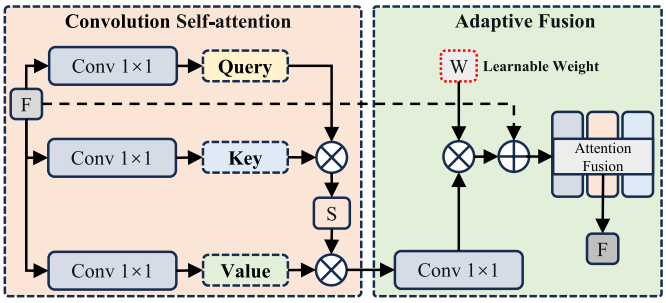


Fig. 6. Illustration of CARM. CARM consists of two stages: convolution self-attention and adaptive fusion. The former is used to refine the more specific semantic information, whereas the latter is used to adaptively incorporate the attention information and update the fused feature.

extraction as be expressed as

$$F_{\text{pre}} = \text{Encoder}\left(\text{Img}_{\text{time1}}\right) \tag{3}$$

$$F_{\text{post}} = \text{Encoder}\left(\text{Img}_{\text{time2}}\right) \tag{4}$$

where $\text{Img}_{\text{time1}}, \text{Img}_{\text{time2}} \in \mathbb{R}^{3\times H\times W}$ represent the input bitemporal images and $F_{\text{pre}}, F_{\text{post}} \in \mathbb{R}^{C\times(H/8)\times(W/8)}$ represent the extracted features of the pretime image and posttime image, respectively.

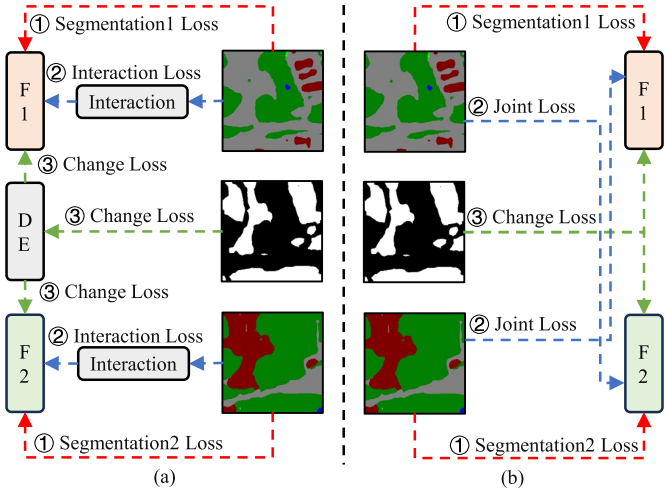


Fig. 7. Optimization path comparison between (b) our strategy and (a) existing strategy. (a) Strategy generally needs to introduce the carefully designed DE module and branch interactions to increase the effective optimization paths. However, in our proposed network, we adopt three structurally unified branches to directly refine the joint bitemporal features, which can keep the same number of optimization paths while avoiding the need for these additional modules.

B. Structurally Unified Triple Branches

As shown in Fig. 4, we adopt three structurally unified branches to refine the specific semantic information for different tasks. Each branch contains two stages: bitemporal information fusion and semantic information refinement.

1) Convolutional Attention Fusion Module: Because the attention mechanism can effectively perceive the feature semantic correlations, we introduce the channel and the spatial attention [41] into CAFM to achieve the bitemporal information fusion. As shown in Fig. 5, because the average pooling and maximum (max) pooling contain different feature information, we set several parallel subbranches to fully capture the useful attention information and further introduce the channel reduction (CR) and spatial fusion (SF) operations to update the incorporated features.

Specifically, for the parallel channel attention module (PCAM), we first deploy the average pooling and max-pooling in channel dimension to extract the feature information. Then, we adopt the multilayer perceptron (MLP) and sigmoid activation to project attention maps and use multiplication to incorporate them. Finally, we add these two incorporated features together and send them into CR to update channel information. For the parallel spatial attention module (PSAM), we first deploy the average pooling and max-pooling in spatial dimension to extract the feature information. Then, we concatenate and send them into three convolution layers with different kernel sizes to project attention maps. After that, we use Sigmoid to normalize them and adopt multiplication to incorporate the attention information into the original feature. Finally, we add these three incorporated features together and send them into SF to update the spatial information. The

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TABLE I
FEATURE DIMENSIONS OF OUR JFRNET IN DIFFERENT STAGES

Encoder	CAFM				CARM				Decoer	
	PCA	CR	PSA	SF	Quary	Key	Value	AF	Class	Difference
[128, 32, 32]	[256, 32, 32]	[256, 32, 32]	[128, 32, 32]	[128, 32, 32]	[1024, 16]	[1024, 16]	[1024, 128]	[128,32,32]	[128,32,32]	[128,32,32]
	-	-	-	-				-	-	[64,32,32]
	[256, 32, 32]	[128, 32, 32]	[128, 32, 32]	[128, 32, 32]				[128,32,32]	[c,256,256]	[1,256,256]

process of CAFM as be expressed as

$$F_{\text{joint}} = \text{Concat}\left(F_{\text{pre}}, F_{\text{post}}\right) \tag{5}$$
$$A^1_{\text{pcam}} = \text{Sigmoid}\left(\text{MLP}\left(\text{AvgPool}\left(F_{\text{joint}}\right)\right)\right) \tag{6}$$
$$A^2_{\text{pcam}} = \text{Sigmoid}\left(\text{MLP}\left(\text{MaxPool}\left(F_{\text{joint}}\right)\right)\right) \tag{7}$$
$$\text{CR}(F) \triangleq \text{ReLU}(\text{BN}(\text{Conv}_{3\times3}(F))) \tag{8}$$
$$F_{\text{pcam}} = \text{CR}\left(A^1_{\text{pcam}} * F_{\text{joint}} + A^2_{\text{pcam}} * F_{\text{joint}}\right) \tag{9}$$
$$F_p = \text{Concat}\left(\text{AvgPool}\left(F_{\text{pcam}}\right), \text{MaxPool}\left(F_{\text{pcam}}\right)\right) \tag{10}$$
$$A^1_{\text{psam}} = \text{Sigmoid}\left(\text{Conv}_{5\times5}\left(F_p\right)\right) \tag{11}$$
$$A^2_{\text{psam}} = \text{Sigmoid}\left(\text{Conv}_{7\times7}\left(F_p\right)\right) \tag{12}$$
$$A^3_{\text{psam}} = \text{Sigmoid}\left(\text{Conv}_{9\times9}\left(F_p\right)\right) \tag{13}$$
$$\text{SF}(F) \triangleq \text{ReLU}(\text{BN}(\text{Conv}_{3\times3}(F))) \tag{14}$$
$$F_{\text{psam}} = \text{SF}\left(\sum_{i=1}^{3} A^i_{\text{psam}} * F_{\text{pcam}}\right) \tag{15}$$

where $F_{\text{joint}} \in \mathbb{R}^{2C\times(H/8)\times(W/8)}$ represents the joint bitemporal image feature, Concat represents the concatenation operation, AvgPool represents the average pooling operation, MaxPool represents the max-pooling operation, $\text{Conv}_{a\times a}$ represents the convolution kernel of size $a \times a$, $*$ represents the element-wise multiplication, $A_{\text{pcam}} \in \mathbb{R}^{2C\times1\times1}$ represents the attention map of PCAM, BN represents the BatchNorm operation, $F_{\text{pcam}} \in \mathbb{R}^{C\times(H/8)\times(W/8)}$ represents the output feature of PCAM, $A_{\text{psam}} \in \mathbb{R}^{1\times(H/8)\times(W/8)}$ represents the attention map of PSAM, and $F_{\text{psam}} \in \mathbb{R}^{C\times(H/8)\times(W/8)}$ represents the output feature of PSAM.

2) Convolutional Attention Refinement Module: As shown in Fig. 4, after CAFM, we introduce the CARM to capture and enhance the more salient semantic information. The detailed structure is shown in Fig. 6. CARM consists of the convolution self-attention module and adaptive fusion module. The former is used to refine the more specific semantic information, which has been widely verified in some CD methods [14], [19], [22], [30]. The latter is used to adaptively incorporate the attention information and update the fused feature. Specifically, we first adopt three convolution layers to, respectively, project the query, key, and value features. Then, we utilize multiplication and Sigmoid activation to model the self-attention information and use one convolution layer to update it. After that, inspired by [35], we set the learnable weight to adaptively incorporate the attention information into the input feature, which can help the model better achieve attention utilization. Finally, we send the incorporated feature into the attention fusion (AF) module to update it. The process of CARM can be expressed as

$$Q = \text{Rep}\left(\text{Conv}_{1\times1}\left(F_{\text{psam}}\right)\right) \tag{16}$$
$$K = \text{Rep}\left(\text{Conv}_{1\times1}\left(F_{\text{psam}}\right)\right) \tag{17}$$
$$V = \text{Rep}\left(\text{Conv}_{1\times1}\left(F_{\text{psam}}\right)\right) \tag{18}$$
$$A_{\text{carm}} = W_{\text{learn}}\text{Conv}_{1\times1}\left(\text{Rep}\left(\text{Softmax}\left(QK^T\right)V\right)\right) \tag{19}$$
$$F_{\text{carm}} = \text{ReLU}\left(\text{BN}\left(\text{Conv}_{3\times3}\left(F_{\text{psam}} + A_{\text{carm}}\right)\right)\right) \tag{20}$$

where Rep represents reshaping the feature dimension, $Q, K \in \mathbb{R}^{((H/8)\times(W/8))\times(C/8)}$, $V \in \mathbb{R}^{((H/8)\times(W/8))\times C}$, respectively, represent the query, key, and value features, $W_{\text{learn}} \in \mathbb{R}^{1\times1}$ represents the learnable weight, $A_{\text{carm}} \in \mathbb{R}^{C\times(H/8)\times(W/8)}$ represents the attention map of CARM, and $F_{\text{carm}} \in \mathbb{R}^{C\times(H/8)\times(W/8)}$ represents the output of CARM.

C. Decoder

As shown in Fig. 4, we utilize three shallow CNN decoders to, respectively, obtain the bitemporal category maps and BCM and further mask bitemporal category maps with BCM to generate the final SCMs. Similar to the previous work [30], the BCM decoder contains two convolution layers to gradually reduce the channel dimension. The bitemporal decoders contain one convolution layer to directly project the features into the category dimension. Because the encoder outputs the 8x downsampled features, we also adopt the bilinear interpolation to upsample the prediction results into the size of the input images. The process of difference information prediction can be expressed as

$$\text{BCM} = \text{ReLU}\left(\text{BN}\left(\text{Conv}_{1\times1}\left(F^{\text{change}}_{\text{carm}}\right)\right)\right) \tag{21}$$
$$\text{BCM} = \text{UP}(\text{Conv}_{1\times1}(BCM)) \tag{22}$$
$$\text{Catg}_{\text{pre}} = \text{UP}\left(\text{Conv}_{1\times1}\left(F^{\text{pre}}_{\text{carm}}\right)\right) \tag{23}$$
$$\text{Catg}_{\text{post}} = \text{UP}\left(\text{Conv}_{1\times1}\left(F^{\text{post}}_{\text{carm}}\right)\right) \tag{24}$$
$$\text{SCM}_{\text{pre}} = \text{Max}(\text{BCM}) * \text{Max}\left(\text{Catg}_{\text{pre}}\right) \tag{25}$$
$$\text{SCM}_{\text{post}} = \text{Max}(\text{BCM}) * \text{Max}\left(\text{Catg}_{\text{post}}\right) \tag{26}$$

where $F^{\text{pre}}_{\text{carm}}$, $F^{\text{post}}_{\text{carm}}$, and $F^{\text{change}}_{\text{carm}}$, respectively, represent the CARM features of the pretime, change, and posttime branches; $\text{BCM} \in \mathbb{R}^{1\times H\times W}$ represents the probability map of the BCM; Catg_{pre} *and* $\text{Catg}_{\text{post}} \in \mathbb{R}^{c\times H\times W}$, respectively, represent the probability maps of the pretime image and posttime image with c types of ground categories; UP represents the bilinear interpolation; Max represents extracting the category serial number with the maximum probability value; and $\text{SCM}_{\text{pre}}, \text{SCM}_{\text{post}} \in \mathbb{R}^{1\times H\times W}$ represent the bitemporal SCMs.

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D. Loss Function and Optimization Analysis

Following the previous studies [42], [43], [44], [45], we choose the cross-entropy loss (CEL) to optimize the classification task of each branch. Specifically, for the bitemporal classification branches, we choose the multiclass CEL. For the BCD branch, we adopt the binary CEL. The details are shown as follows:

$$\mathrm{CEL}(\mathrm{Pre}, \mathrm{GT}) \triangleq -\frac{1}{N}\sum_{i=1}^{N} \mathrm{GT}_i \log(\mathrm{Pre}_i) \tag{27}$$

$$\mathrm{Loss}_{b1} = \mathrm{CEL}\left(\mathrm{Pre}_{\mathrm{catg1}}, \mathrm{GT}_{\mathrm{catg1}}\right) \tag{28}$$

$$\mathrm{Loss}_{b2} = \mathrm{CEL}\left(\mathrm{Pre}_{\mathrm{catg2}}, \mathrm{GT}_{\mathrm{catg2}}\right) \tag{29}$$

$$\mathrm{Loss}_{bcd} = -\mathrm{GT}_c \log(\mathrm{Pre}_c) - (1-\mathrm{GT}_c)\log(1-\mathrm{Pre}_c) \tag{30}$$

where GT_i and Pre_i represent the ground truth and the predicted probability maps of the ith category, respectively, N represents the total category number, $\mathrm{Pre}_{\mathrm{catg}}, \mathrm{GT}_{\mathrm{catg}}$ represent the predicted and the ground-truth category maps, respectively, $\mathrm{Loss}_{b1}, \mathrm{Loss}_{b2}$ represent the CEL loss of pretime and posttime classification branches, respectively, $\mathrm{Pre}_c, \mathrm{GT}_c$ represent the predicted and the ground-truth BCMs, respectively, and Loss_{bcd} represents the CEL of BCD branch.

Besides, we also introduce the SCLoss [30] to connect the bitemporal SCMs with BCM. Specifically, SCLoss uses the cosine similarity function to optimize the distance of bitemporal category maps, where the similarity value should be lower in the changed regions and higher in the unchanged regions

$$\mathrm{Loss}_{sc} = \begin{cases} 1-\cos\left(\mathrm{Pre}_1^{i,j}, \mathrm{Pre}_2^{i,j}\right), & \mathrm{GT}_{\mathrm{BCM}}^{i,j} = 0 \\ \cos\left(\mathrm{Pre}_1^{i,j}, \mathrm{Pre}_2^{i,j}\right), & \mathrm{GT}_{\mathrm{BCM}}^{i,j} = 1 \end{cases} \tag{31}$$

$$\mathrm{Loss}_{\mathrm{total}} = (\mathrm{Loss}_{b1} + \mathrm{Loss}_{b2}) * 0.5 + \mathrm{Loss}_{cd} + \mathrm{Loss}_{sc} \tag{32}$$

where $\mathrm{Pre}_1^{i,j}, \mathrm{Pre}_2^{i,j}$ represent the predicted probability vectors of bitemporal category maps in position (i, j), $\mathrm{GT}_{\mathrm{BCM}}^{i,j}$ represents the ground-truth label of BCM in position (i, j), and cos represents the cosine similarity function. To sum up, the total loss of our JFRNet can be expressed as (32).

Peculiarly, the optimization paths of the loss function can also verify the effectiveness of our proposed strategy. Specifically, because both the category information and the change information are derived from bitemporal images, all of the loss optimization information will finally be fed back to the bitemporal features. As shown in Fig. 7(a), strategy (a) generally needs to introduce the carefully designed DE module and branch interactions to increase the effective optimization paths, which increases the complexity of the semantic information utilization. As a result, the total optimization targets of strategy (a) $\mathrm{Opt}_{\mathrm{total}}^{a}$ can be summarized as (33), where $\mathrm{Opt}_{\mathrm{seg}}$ represents optimizing the segmentation branch, Opt_{DE} represents optimizing the DE module, $\mathrm{Opt}_{\mathrm{inter}}$ represents optimizing the interaction module, and $\oplus$ represents adding the optimization target

$$\mathrm{Opt}_{\mathrm{total}}^{a} = \mathrm{Opt}_{\mathrm{seg1}} \oplus \mathrm{Opt}_{\mathrm{seg2}} \oplus \mathrm{Opt}_{DE} \oplus \mathrm{Opt}_{\mathrm{inter}} \tag{33}$$

$$\mathrm{Opt}_{\mathrm{total}}^{b} = \mathrm{Opt}_{\mathrm{ref1}} \oplus \mathrm{Opt}_{\mathrm{ref2}} \oplus \mathrm{Opt}_{\mathrm{ref3}}. \tag{34}$$

Differently, in our proposed method, we adopt three structurally unified branches to directly refine the joint bitemporal features, which can keep the same number of optimization paths while avoiding the need for these additional modules. Therefore, the total optimization targets of our strategy $\mathrm{Opt}_{\mathrm{total}}^{b}$ can be summarized as (34), where $\mathrm{Opt}_{\mathrm{ref}}$ represents optimizing the structurally unified refinement branch.

E. Model Details

Considering the limitation of computing resources, we unify the input image size to 256 × 256, which has already been widely adopted by the previous studies [19], [46], [47] (see Section IV-B for detail information). Table I reports the feature dimensions of our JFRNet in different stages, where $[C, H, W]$ represent the feature channel, height and width, respectively, $[L, C]$ represent the feature-length and channel, respectively, the first row is the input feature dimension, the second row is the middle feature dimension, the last row is the output feature dimension, PCA represents the parallel channel attention mechanism, PSA represents the parallel spatial attention mechanism, AF represents the adaptive fusion module, Class and Difference represent the classification branch decoder and BCD branch decoder, respectively, and c represents the category number of SCM. We first utilize ResNet-34 [34] to obtain the 8× downsampled image feature whose channel dimension increases to 128. Then, we concatenate the bitemporal image features in the channel dimension and send them into CAFM to achieve bitemporal information fusion which finally outputs the feature of size [128,32,32]. After that, we utilize CARM to refine the semantic information which keeps the same feature dimension as the input. Peculiarly, when calculating the self-attention, we flatten the feature height and width to obtain the length dimension. Finally, we obtain the probability maps of BCM and bitemporal category maps by projecting the feature channel dimension. Particularly, the decoder of the difference branch reduces the feature channel to 64 as the middle feature.

IV. Experiment

We verify the effectiveness of the proposed method on three available SCD datasets: SECOND [33], Landsat-SCD [27], and HRSCD [31]. Table II reports the detail information of these three datasets. Tables III–VI display the quantitative results of our JFRNet and the other advanced CD methods. Fig. 8 illustrates the visualized SCD process of our method. Figs. 9–11 report the detail performance differences between our model and the baseline methods. Figs. 12–14 display some visualized examples. Tables VII–IX and Fig. 15 report the ablation results.

A. Compared Methods

Eleven representative CD methods are totally selected to make the comparisons. Specifically, FC-EF [7], FC-SC [7], and FC-SD [7] modify UNet [36] to achieve CD. DTSCN [10] adopts both the BCD and segmentation task to constrain the model. BIT [14] introduces the self-attention mechanism of the transformer [37] to refine the bitemporal image features. HRSCD-str2 [31] adopts the single-branch structure to directly

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TABLE II
DATE PREPROCESSING OF THESE THREE SCD DATASETS

Dataset	Category	Patch Size	Train	Val	Test
SECOND [33]	7	256 × 256	8,310	1,188	2,374
Landsat-SCD [27]	5	256 × 256	6,678	954	1,908
HRSCD [31]	6	256 × 256	5,735	820	1,639

predict the SCM. Both HRSCD-str3 [31] and HRSCD-str4 [31] adopt the triple-branch structure to achieve SCD. However, the difference is that these three branches are independent in HRSCD-str3 but connected in HRSCD-str4. SCDNet [29] adopts two UNet-style branches to predict SCMs. SSCDL [30] introduces a new triple-branch process to achieve SCD. Bi-SRNet [30] further introduces the attention mechanism to achieve branch interactions. Peculiarly, as mentioned in Section I, when keeping the model structure unchanged, the single-branch CD methods generally need to consider all of the "from–to" semantic category transitions, which will badly increase the prediction difficulty. Therefore, considering the fairness and feasibility, we just add one additional shallow classifier into these single-branch methods to directly predict bitemporal SCMs, which can bring the minimum structure influences while keeping the effective learning. In our experiments, we take SSCDL and Bi-SRNet as our baseline methods.

B. Datasets

Table II reports the data preprocessing of our experiments on these three SCD datasets.

The SECOND [33] dataset has 2968 pairs of available bitemporal images of size 512 × 512. Based on changed regions, it totally annotates six categories of ground objects: water, ground, low vegetation, trees, buildings, and sports fields. Considering the limitation of computation resources, we cut each image into four smaller nonoverlapping patches of size 256 × 256. After that, we totally obtained 11 872 pairs of samples and randomly divided them with the ratio of 7 : 1 : 2 for training, validation, and testing.

The Landsat-SCD [27] dataset has 8468 pairs of available bitemporal images of size 416 × 416. Based on changed regions, it totally annotates four categories of ground objects: farmland, desert, building, and water. To reduce the data leakage as much as possible while keeping the effective model learning, we first remove the manually enhanced samples to obtain 2385 pairs of completely different images. Then, we cut each sample into four smaller patches of size 256 × 256 by setting the overlap step to 96. Finally, we randomly divide the patches with the ratio of 7 : 1 : 2 for training, validation, and testing.

The HRSCD [31] dataset has 291 pairs of available bitemporal images of size 10000 × 10000. Based on changed regions, it totally annotates five categories of ground objects: artificial surface, agricultural area, forest, wetland, and water. In our experiments, considering the limitation of computation resources, we select 53 pairs of original images that contain abundant semantic category information. Similarly, we cut each image into 1521 smaller nonoverlapping patches of size 256 × 256 and further filter out all of the useless images to obtain 8194 effective samples. Finally, we also randomly divide them with the ratio of 7 : 1 : 2 for training, validation, and testing.

Particularly, to achieve fair comparisons, all of the methods adopt the same samples for experiments. We repeat the key experiments and take the average values as the final metrics to make sure the results are reliable.

C. Evaluation Metrics

In our experiments, we select three metrics to evaluate the model performance: SeK (separated kappa), Fscd, and Avg. Sek is introduced by Yang et al. [33], which only considers the categories of the changed regions to measure the classification performance

$$\rho = \sum_{i=1}^{N} c_{ii} \div \left(\sum_{i=0}^{N} \sum_{j=0}^{N} c_{ij} - c_{00} \right) \tag{35}$$

$$\eta = \sum_{i=0}^{N} (c_{i+} \cdot c_{+i}) \div \left(\sum_{i=0}^{N} \sum_{j=0}^{N} c_{ij} - c_{00} \right)^2 \tag{36}$$

$$\text{Sek} = e^{(\text{IoU}_c - 1)} \cdot (\rho - \eta)/(1 - \eta) \tag{37}$$

where c_{i+}, c_{+i} represent the row summation and column summation of the confusion matrix without c_{00}, respectively.

Fscd is introduced by [30], which is derived from the $F1$ score to balance the classification precision and recall of changed regions

$$P_{\text{scd}} = \sum_{i=1}^{N} c_{ii} \div \sum_{i=1}^{N} \sum_{j=0}^{N} c_{ij} \tag{38}$$

$$R_{\text{scd}} = \sum_{i=1}^{N} c_{ii} \div \sum_{i=0}^{N} \sum_{j=1}^{N} c_{ij} \tag{39}$$

$$F_{\text{scd}} = \frac{2 * P_{\text{scd}} * R_{\text{scd}}}{P_{\text{scd}} + R_{\text{scd}}} \tag{40}$$

where P_{scd} and R_{scd} represent the precision and the recall, respectively.

Avg is the weighted sum of mean intersection over union (mIoU) and overall accuracy (OA). Specifically, mIoU is the average value of the changed IoU and unchanged IoU, which can indicate the BCD performance [48]. OA is the overall proportion of the correctly predicted pixel number, which has been widely applied in BCD [49], [50], [51]. However, because both mIoU and OA consider the unchanged category which has a large proportion and can dominate the metric value, we adopt the weighted sum strategy in the previous SCD work [32] to obtain new comprehensive metrics (Avg) that can better reflect the comprehensive SCD performance

$$\text{IoU}_{uc} = c_{00} \div \left(\sum_{i=0}^{N} c_{i0} + \sum_{j=0}^{N} c_{0j} - c_{00} \right) \tag{41}$$

$$\text{IoU}_c = \sum_{i=1}^{N} \sum_{j=1}^{N} c_{ij} \div \left(\sum_{i=0}^{N} \sum_{j=0}^{N} c_{ij} - c_{00} \right) \tag{42}$$

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TABLE III
QUANTITATIVE EXPERIMENTAL RESULTS ON THREE AVAILABLE SCD DATASETS

Method	Structure Type	SECOND [33]			Landsat-SCD [27]			HRSCD [31]		
		Sek (%)	Fscd (%)	Avg (%)	Sek (%)	Fscd (%)	Avg (%)	Sek (%)	Fscd (%)	Avg (%)
FC-EF [7]	Single Branch	3.239	40.214	70.545	14.556	61.559	77.639	2.426	37.105	65.048
FC-SC [7]	Single Branch	5.451	43.268	72.097	18.037	65.002	79.044	6.472	49.245	69.040
FC-SD [7]	Single Branch	4.590	38.887	70.737	13.130	59.964	77.293	4.945	46.136	67.965
DTSCN [10]	Single Branch	18.688	58.797	78.364	31.885	75.718	84.562	16.411	64.846	75.176
BIT [14]	Single Branch	19.651	59.993	78.691	40.662	79.979	87.131	17.323	64.982	75.619
HRSCD-str2 [31]	Single Branch	12.458	51.857	76.226	33.694	76.551	85.291	18.312	66.492	76.149
HRSCD-str3 [31]	Triple Branch	13.692	53.961	76.099	35.470	75.994	85.717	7.798	52.501	72.905
HRSCD-str4 [31]	Triple Branch	20.186	59.740	79.609	40.140	78.892	87.237	10.299	54.651	73.523
SCDNet [29]	Dual Branch	20.519	60.840	79.567	-	-	-	8.401	52.821	72.535
SSCDL [30]	Triple Branch	21.149 $*$	61.786 $*$	80.084 $*$	44.279 $*$	82.092 $*$	88.283 $*$	18.148 $*$	66.228 $*$	75.980 $*$
Bi-SRNet [30]	Triple Branch	22.247 $*$	62.149 $*$	80.082 $*$	42.954 $*$	81.478 $*$	87.832 $*$	18.486 $*$	66.482 $*$	75.928 $*$
JFRNet (ours)	Triple Branch	**22.562** $*$	**62.631** $*$	**80.095** $*$	**46.394** $*$	**83.488** $*$	**88.755** $*$	**18.778** $*$	**66.655** $*$	**76.204** $*$

- represents that the compared metrics are not available. $*$ represents that we repeat the experiments and take the average values as the final metrics.

TABLE IV
TECHNIQUE COMPARISON BETWEEN OUR JFRNET AND THE BASELINE METHODS

Method	Backbone	Computation Costs		Triple Branch	
		Params (M)	MACs (G)	DE	Interaction
SSCDL [30]	ResNet-34 [34]	23.313	47.440	✓	✗
Bi-SRNet [30]	ResNet-34 [34]	23.375	47.536	✓	✓
JFRNet (ours)	ResNet-34 [34]	23.295	47.378	✗	✗

Params: model parameters, MACs: multiply-accumulate operations.

TABLE V
QUANTITATIVE CROSS-DATASET TEST RESULTS

Method	Fine-tuning Targets	Fscd (%)	Avg (%)
SSCDL [30]	Semantic Classifiers, One Epoch	50.251	69.204
Bi-SRNet [30]	Semantic Classifiers, One Epoch	52.718	68.506
JFRNet (ours)	Semantic Classifiers, One Epoch	**56.643**	**69.241**

$$\text{mIoU} = (\text{IoU}_c + \text{IoU}_{uc})/2 \tag{43}$$

$$\text{OA} = \sum_{i=0}^{N} c_{ii} \div \sum_{i=0}^{N}\sum_{j=0}^{N} c_{ij} \tag{44}$$

$$\text{Avg} = \text{mIoU} * 0.5 + \text{OA} * 0.5 \tag{45}$$

where c_{ij} represents the pixel number of the confusion matrix [52], [53] in position (i, j) (the predicted category is "i" but the ground truth is "j"), IoU_c represents the changed IoU, and IoU_{uc} represents the unchanged IoU.

D. Implementation Details

We implement our experiments on the Pytorch framework [54]. To achieve fair comparisons, all selected methods adopt the same training strategy. Specifically, the optimizer is stochastic gradient descent (SGD) with Nesterov momentum. The training data augmentation strategy includes random flipping and rotating. The device is one NVIDIA TITAN RTX 24G GPU. The total number of epochs is set to 50. The initial learning rate is 0.1 and decays with the iterations as $0.1 * (1 - \text{interactions/total_interactions})^{1.5}$. We utilize the validation set to evaluate the model and adopt the best model parameter to test.

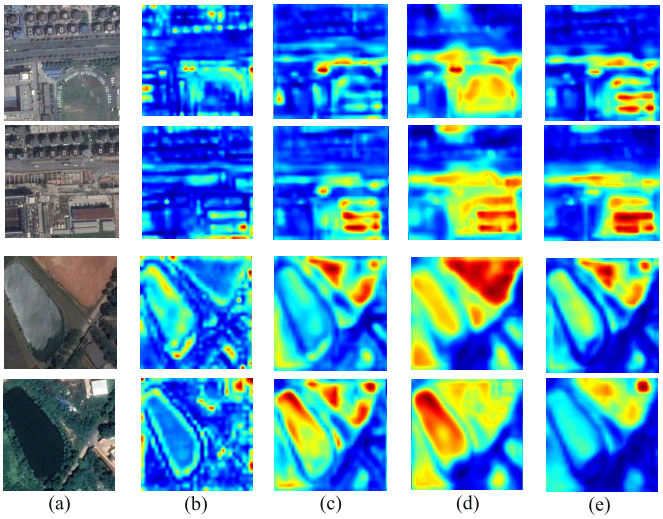


Fig. 8. Visualization SCD process of JFRNet. (a) Bitemporal images. (b) Extracted bitemporal features. (c) Bitemporal CAFM features. (d) Bitemporal CARM features. (e) (Top) BCD branch features of CAFM and (bottom) CARM.

E. Experimental Analysis

Table III reports the quantitative metrics of our proposed method and the other compared methods on these three available SCD datasets. Compared with FC-EF and FC-SD, FC-SC overall obtains the more advanced SCD performance, which indicates that the joint feature is beneficial for SCD. Because BIT introduces the attention mechanism of the transformer to capture the spatiotemporal correlations of bitemporal images, its performance is the highest among the modified BCD

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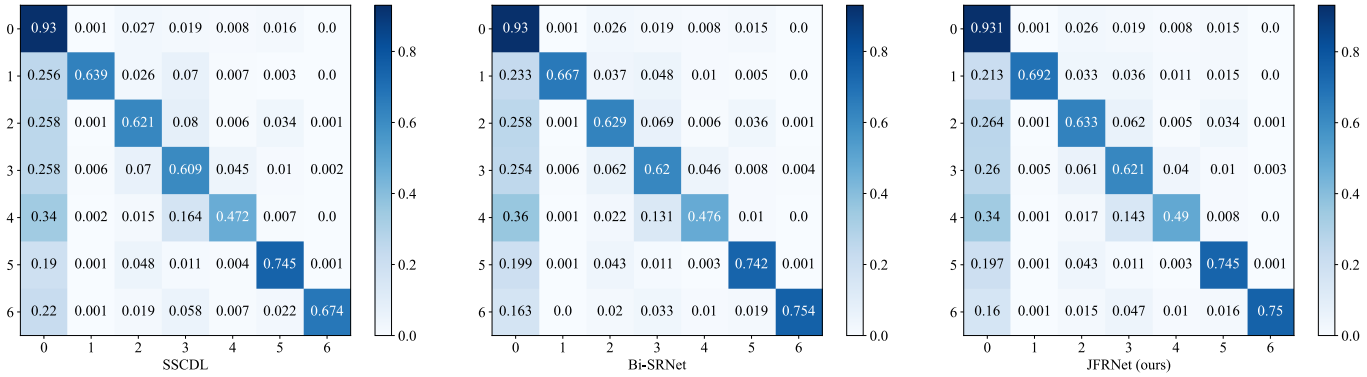


Fig. 9. Confusion matrix comparison between our JFRNet and the baseline methods (SSCDL and Bi-SRNet). 0–6, respectively, represent "unchanged," "water," "ground," "low vegetation," "tree," "building," and "sports field."

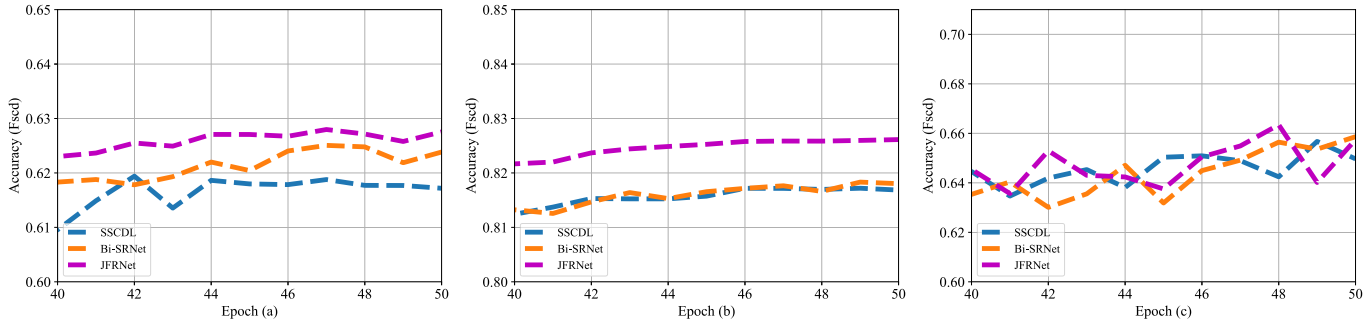


Fig. 10. Validation accuracy curves of our JFRNet and the baseline methods on these three SCD datasets. The curves of (a)–(c), respectively, come from the SECOND, Landsat-SCD, and HRSCD datasets.

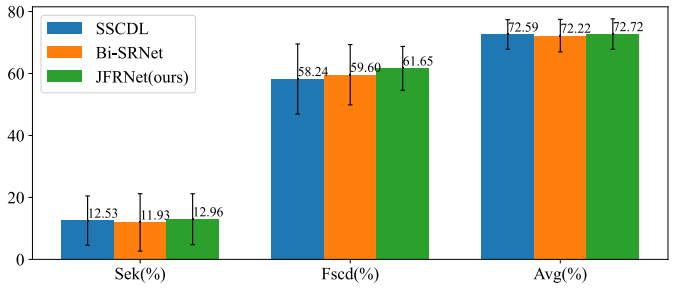


Fig. 11. Performance fluctuation comparison between our JFRNet and the baseline methods. We calculate the mean and standard deviation values of these three metrics before and after the cross-dataset validation.

TABLE VI
QUANTITATIVE EXPERIMENT RESULTS ON THE ISOLATED LANDSAT-SCD DATASET

Method	Landsat-SCD Isolated Test		
	Sek (%)	Fscd (%)	Avg (%)
FC-EF [7]	15.925	62.567	78.266
FC-SC [7]	19.602	66.093	79.677
FC-SD [7]	14.457	61.100	77.895
DTSCN [10]	33.844	76.757	85.187
BIT [14]	41.921	80.542	87.494
HRSCD-str2 [31]	35.666	77.578	85.909
HRSCD-str3 [31]	37.564	77.143	86.391
HRSCD-str4 [31]	42.334	79.960	87.888
SSCDL [30]	45.521 ∗	82.629 ∗	88.598 ∗
Bi-SRNet [30]	44.177 ∗	82.003 ∗	88.154 ∗
JFRNet (ours)	**47.465** ∗	**83.896** ∗	**89.038** ∗

∗ represents that we repeat the experiments and take the average values as the final metrics.

methods. Although HRSCD-str3 and HRSCD-str4 lag behind the HRSCD-str2 on HRSCD datasets, they perform better on the SECOND and Landsat-SCD datasets, which indicates that the triple-branch structure is overall more advanced than the single-branch structure. Similarly, the performance gap between HRSCD-str3 and HRSCD-str4 shows that the branch interaction does improve the model performance. Besides, although SCDNet exceeds the most compared methods on the SECOND dataset, its performance dramatically decreases on the HRSCD dataset. Differently, because SSCDL improves the triple-branch structure and Bi-SRNet further introduces the branch interaction, they obtain more advanced SCD performance.

However, compared with all of the selected methods, our proposed method achieves the best accuracy. For example, the Sek of our JFRNet on these three datasets, respectively, exceeds the advanced Bi-SRNet by 0.315, 3.44, and 0.292 points. Particularly, as mentioned in Section IV-B, because we cut the Landsat-SCD dataset in an overlapping way to ensure the compared methods can effectively converge, we also test the metrics of these methods on the whole isolated subset to reduce the influences brought by the data leakage. The quantitative experiment results in Table VI indicate that our model still obtains the highest performance. Besides, we further repeat the experiments of our JFRNet and the baseline methods (SSCDL and Bi-SRNet), which can statistically prove the effectiveness of our method.

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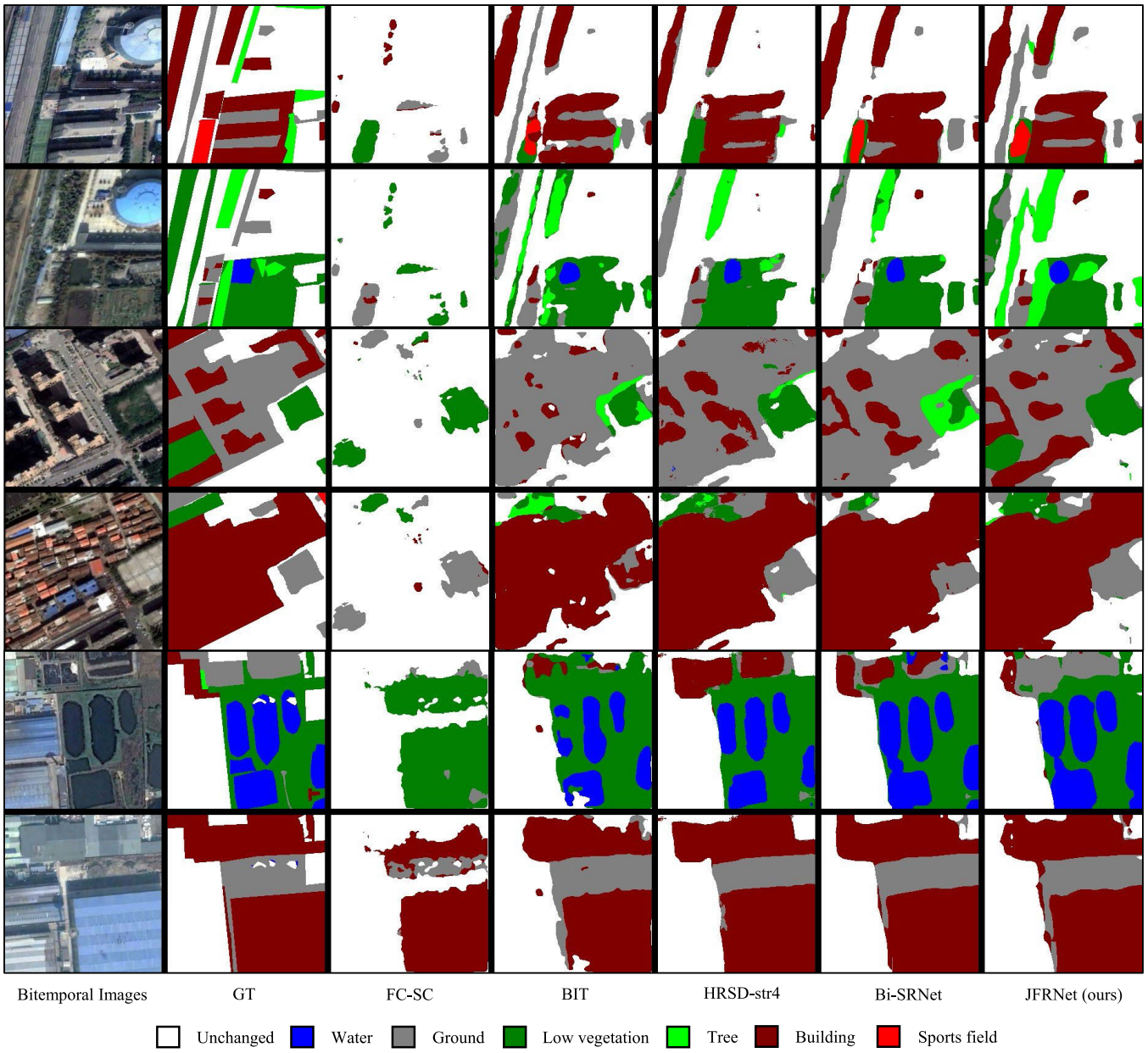


Fig. 12. Some visualized examples on the SECOND dataset.

Furthermore, to achieve a more comprehensive comparison with the baseline methods, we also display the model usage techniques, confusion matrices, validation curves, and cross-dataset metrics. Tables IV and V and Figs. 9–11 report the relevant details. 1) For the usage techniques, we modify the triple-branch structure and process while adopting the same backbone, which can fairly reflect the effectiveness of the proposed improvements. As the results in Tables III and IV show, when keeping similar computation costs, our JFRNet can achieve better performance improvement while avoiding the need for both the specially designed DE module and the branch interactions. 2) For the confusion matrices, as the result in Fig. 9 shows, the "unchanged" category accuracy of our JFRNet is the highest, which indicates that our model can perform the change task more precisely. Although Bi-SRNet performs better in the classification of the "sports field," our accuracy in the rest five ground categories is still the highest, which indicates that JFRNet can overall perform the segmentation task more precisely. 3) For the validation curves, as the result in Fig. 10 shows, the validation accuracy of our method is higher than SSCDL and Bi-SRNet on both the SECOND and Landsat-SCD datasets. Although the curves do fluctuate on the HRSCD dataset, our JFRNet still achieves the highest metric in epoch 48. 4) For the cross-dataset performance, we directly introduce the model trained on Landsat-SCD into HRSCD to obtain cross-dataset metrics. Specifically, considering the ground categories between Landsat-SCD and HRSCD are quite different, we first utilize the parameters trained on Landsat-SCD except the semantic classifiers to initialize the model and then fine-tune them just through one epoch on HRSCD to obtain the quantitative results. As shown in Table V, although the metrics do decrease

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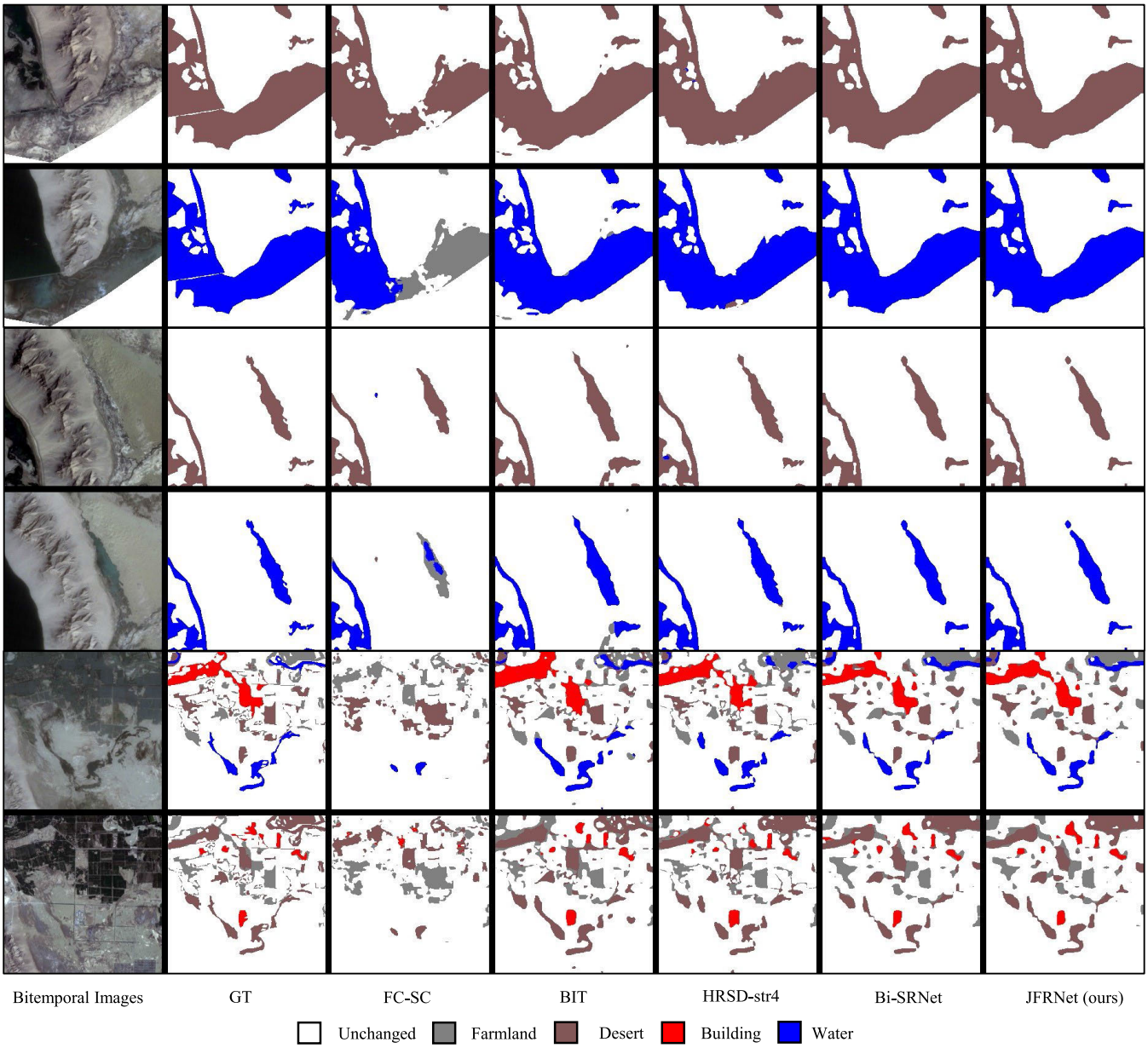


Fig. 13. Some visualized examples on the Landsat-SCD dataset.

compared with Table III, our method still performs better than the baseline methods. Peculiarly, Fig. 11 further intuitively displays the performance fluctuation before and after the cross-dataset validation. Obviously, the fluctuations of SSCDL or Bi-SRNet on these three metrics are always higher than our proposed method.

Fig. 8 illustrates the visualized SCD process of our method. As shown in Fig. 8(b), although the bitemporal image features obtain the preliminary semantic information, the interest change regions are not significant and many pseudo-changes are also extracted. Then, we utilize CAFM to achieve the bitemporal information fusion. As shown in Fig. 8(c) and (e), the interest semantic regions become more clear but they still lack the complete boundary information. Therefore, we introduce CARM to further capture and enhance the more salient semantic information. As shown in Fig. 8(d) and (e), the changed regions are very salient and the pseudo-change regions are also effectively suppressed.

Figs. 12–14 display some visualized results of our proposed method and the other compared methods on these three available SCD datasets. Although the gap does exist between the prediction and GT results, our method still performs best both in changed region detection and classification. For example, for the first group results in Fig. 12, although BIT obtains the more precise changed regions, its classification results are behind ours. For the second group results in Fig. 12, many methods misclassify the low vegetation area as trees, while our JFRNet can correctly recognize them. For the last group results in Fig. 12, many methods misidentify the ground as buildings, while our method can better distinguish them. For the first and second group results in Fig. 13, although BIT, HRSCD-str4, and Bi-SRNet also correctly predict the categories, our

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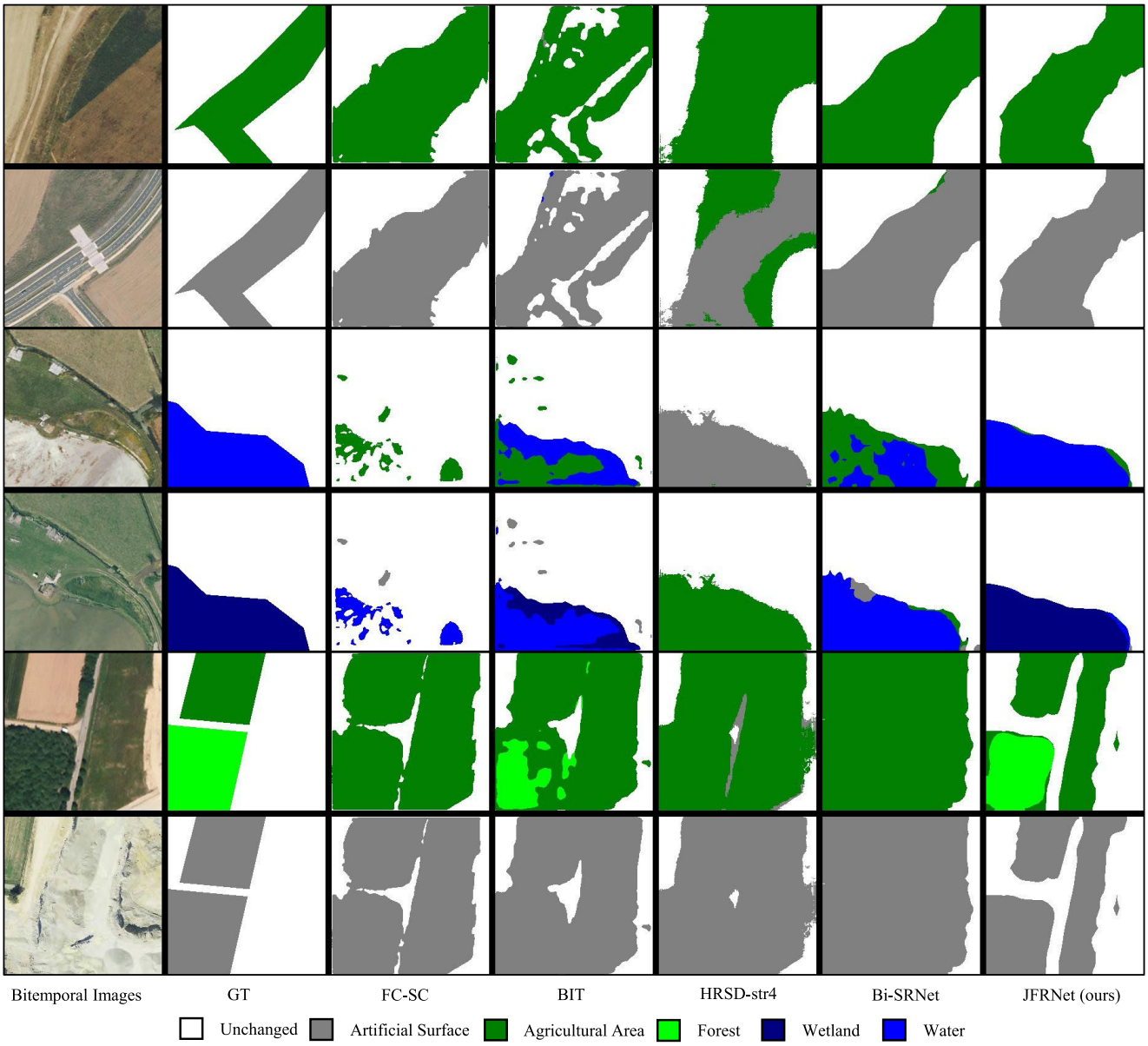


Fig. 14. Some visualized examples on the HRSCD dataset.

changed boundaries are closer to GT. For the last group results in Fig. 13, our JFRNet obtains the more complete regions in the changed water and building. For the first group results in Fig. 14, although all of the selected methods obtain similar bitemporal categories, our predicted change information is still closer to GT. For the second and last group results in Fig. 14, the SCMs of our JFRNet have far more precise information both in change and classification tasks.

To sum up, various quantitative and qualitative experiment results on these three available SCD datasets fully verify the effectiveness of our method. We think this can be attributed to two aspects. 1) For the SCD process, we take the joint bitemporal image features as the unified input, which can help each branch perceive the bitemporal semantic correlations without any additional branch interaction operations. 2) For the SCD structure, we introduce CAFM and CARM to construct each branch, which can help our model extract precise semantic information without the specially designed DE block.

F. Ablation Study

We perform ablation studies for two aspects: module necessity and module details. Besides, all of the ablation metrics are the average values after the repeated experiments, which can statistically verify the proposed improvements.

Table VII reports the quantitative ablation results on the module necessity. In our proposed JFRNet, we introduce CAFM consisting of PCAM, PSAM, and CARM to fuse and refine the joint bitemporal image features, respectively. Therefore, we solely remove them to verify the necessity. Particularly, to test the effectiveness of the parallel subbranches, we also replace PCAM and PSAM with the original attention modules in [41]. The experimental metrics indicate that our proposed modules do bring benefits. For example, without CAFM and CARM, the Sek decreases by 0.32 and 0.746 points, respectively. Table VIII reports the quantitative ablation results on the convolution kernel size.

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TABLE VII
ABLATION STUDIES ON THE MODULE NECESSITY

CAFM		CARM	Sek (%)	Fscd (%)	Avg (%)
PCAM	PSAM				
✗	✗	✓	22.242 ∗	62.271 ∗	79.833 ∗
☒	☒	✓	21.570 ∗	61.662 ∗	79.776 ∗
✗	✓	✓	22.495 ∗	62.443 ∗	79.994 ∗
✓	✗	✓	22.326 ∗	62.440 ∗	79.884 ∗
✓	✓	✗	21.816 ∗	61.928 ∗	79.677 ∗
✓	✓	✓	**22.562** ∗	**62.631** ∗	**80.095** ∗

☒ represents we adopt the original attention modules in [41].
∗ represents that we repeat the experiments and take the average values as the final metrics.

TABLE VIII
ABLATION STUDIES ON THE CONVOLUTION KERNEL SIZE

CAFM		CARM		
Conv1	Conv3	Conv5+Conv7+Conv9	Conv7+Conv9+Conv11	Fscd (%)
✓	✗	✓	✗	62.591 ∗
✗	✓	✗	✓	61.137 ∗
✗	✓	✓	✗	**62.631** ∗

∗ represents that we repeat the experiments and take the average values as the final metrics.

Specifically, in CAFM, we explore the convolution of CR and SF. In CARM, we explore the convolution of the three parallel subbranches. The experimental metrics indicate that the Conv3 and the Conv5,7,9 are more suitable. We think the possible reason is that the other convolution kernels are too small (such as Conv1) or too large (such as Conv11) for our model.

Table IX reports the quantitative ablation results on the upsampling strategy. As mentioned in Section III-C, because the encoder outputs the 8x downsampled features, we also need to adopt the upsampling operation to recover the prediction results into the size of the input images. Specifically, we verify two feasible strategies: bilinear interpolation and convolution. The experimental metrics indicate that the former is more suitable for our model. We think the possible reason is that the output feature of CARM already has high-level semantic information which makes it difficult for the shallow convolution layers to directly achieve the upsampling in a learnable way. Fig. 15 displays some visualized ablation results. Obviously, CAFM and CARM can help our model obtain more precise results. For example, for the first and last group results, adopting these two proposed modules has better performance on the BCD task. For the second group results, the model with both can classify the changed area more accurately.

V. DISCUSSION

In our experiments, although the quantitative metrics and visualized examples do verify the advancement of our proposed method, they still reveal some detail problems. 1) For the selected three evaluation metrics in Table III, the improvements of Sek and Fscd are more effective than Avg.

TABLE IX
ABLATION STUDIES ON THE UPSAMPLING STRATEGY

Upsampling		Sek (%)	Fscd (%)	Avg(%)
Convolution	Bilinear Interpolation			
✓	✗	21.779 ∗	61.759 ∗	79.619 ∗
✗	✓	**22.562** ∗	**62.631** ∗	**80.095** ∗

∗ represents that we repeat the experiments and take the average values as the final metrics.

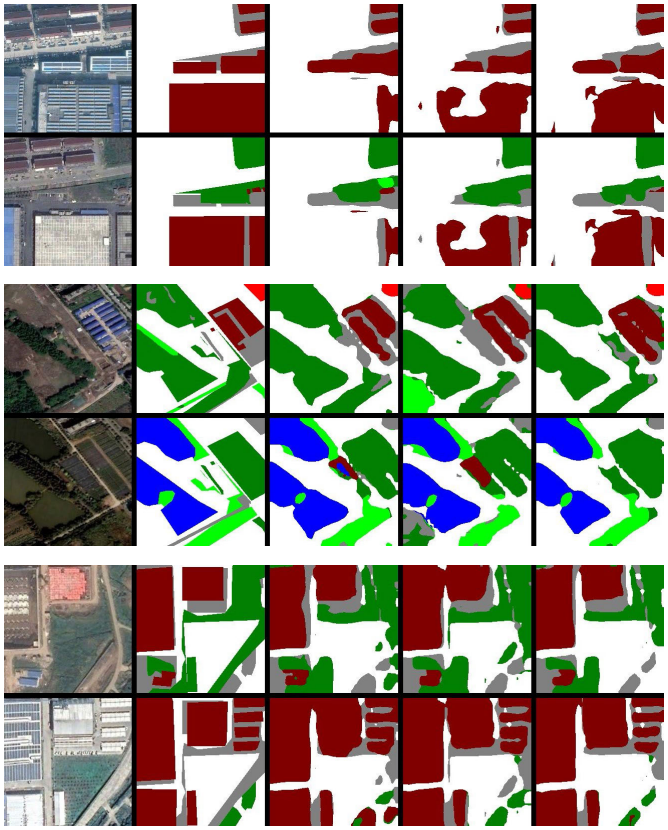


Fig. 15. Some visualized examples on the ablation study. WO represents without.

We think the possible reason is that Avg considers the unchanged category (C_{00}) which has a large pixel proportion and can dominate the metric values. 2) For the three available datasets in our experiments, the quantitative performance gap between our proposed method and the benchmark Bi-SRNet on Landsat-SCD is higher than on SECOND and HRSCD. We think the possible reason is that the changed regions in Landsat-SCD have more complex boundaries and are closer to the background, which can better differentiate the model performance. 3) For the visualized examples in Fig. 12, it is difficult for models to distinguish the difference between low vegetation and trees. We think the possible reason is that these two categories are highly similar in color and texture which are difficult for SCD models to classify. 4) For the visualized examples in Fig. 13, BIT performs better than ours in detecting the line-style changed regions. We think the possible reason is that BIT is a single-branch method that has natural advantages in changed region detection. 5) For the cross-dataset validation

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results in Table V, the existing SCD models still show high performance fluctuation. We think the possible reason is that the existing SCD datasets have great differences in semantic categories, which makes models difficult to deal with the ground categories that have never been trained.

Therefore, how to help SCD models mine the more precise bitemporal information is still worth more exploration. Recently, the graph-based methods that have achieved success in heterogenous CD bring us new inspiration. Specifically, many studies [55], [56], [57], [58] indicate that the graph can efficiently represent the bitemporal structure information without any additional supervision information. Compared with the image pixels, the ground structure has the characteristic of imaging modality-invariant and is more robust to the imaging conditions. Particularly, Sun et al. [57] further successfully demonstrate the effectiveness of the structure consistency graph in homogeneous CD. Therefore, in our future research, we can attempt to utilize the graph to capture the structure-based bitemporal information and try to integrate it into the triple-branch structure to help SCD models obtain more precise results.

VI. Conclusion

In this article, we propose a new network named JFRNet for bitemporal remote-sensing image SCD. Extensive experimental results on three available datasets indicate that our proposed JFRNet successfully simplifies the reasoning process and obtains better SCD performance.

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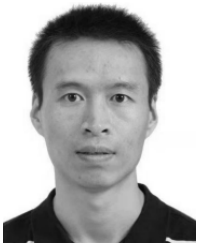
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